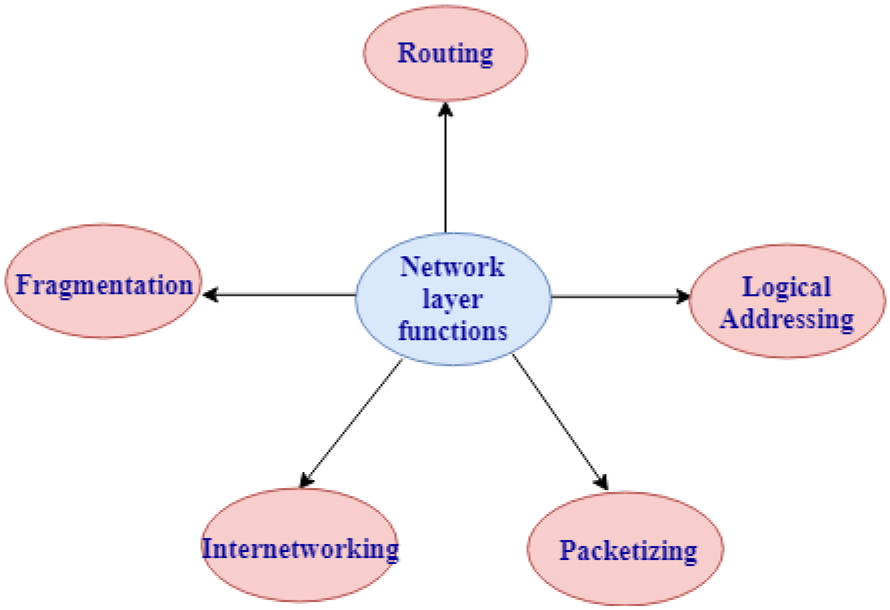

- Routing: The network layer determines the best route from source to destination. This function is known as routing.
- Logical addressing: The network layer defines the addressing scheme to identify each device uniquely.
- Packetizing: The network layer receives the data from the upper layer and converts the data into packets. This process is known as packetizing.
- Internetworking: The network layer provides the logical connection between the different types of networks for forming a bigger network.
- Fragmentation: It is a process of dividing the packets into the fragments.

4. Transport Layer

- It delivers the message through the network and provides error checking so that no error occurs during the transfer of data.
- It provides two kinds of services:
    - Connection-oriented transmission: In this transmission, the receiver sends the acknowledgement to the sender after the packet has been received.
    - Connectionless transmission: In this transmission, the receiver does not send the acknowledgement to the sender.